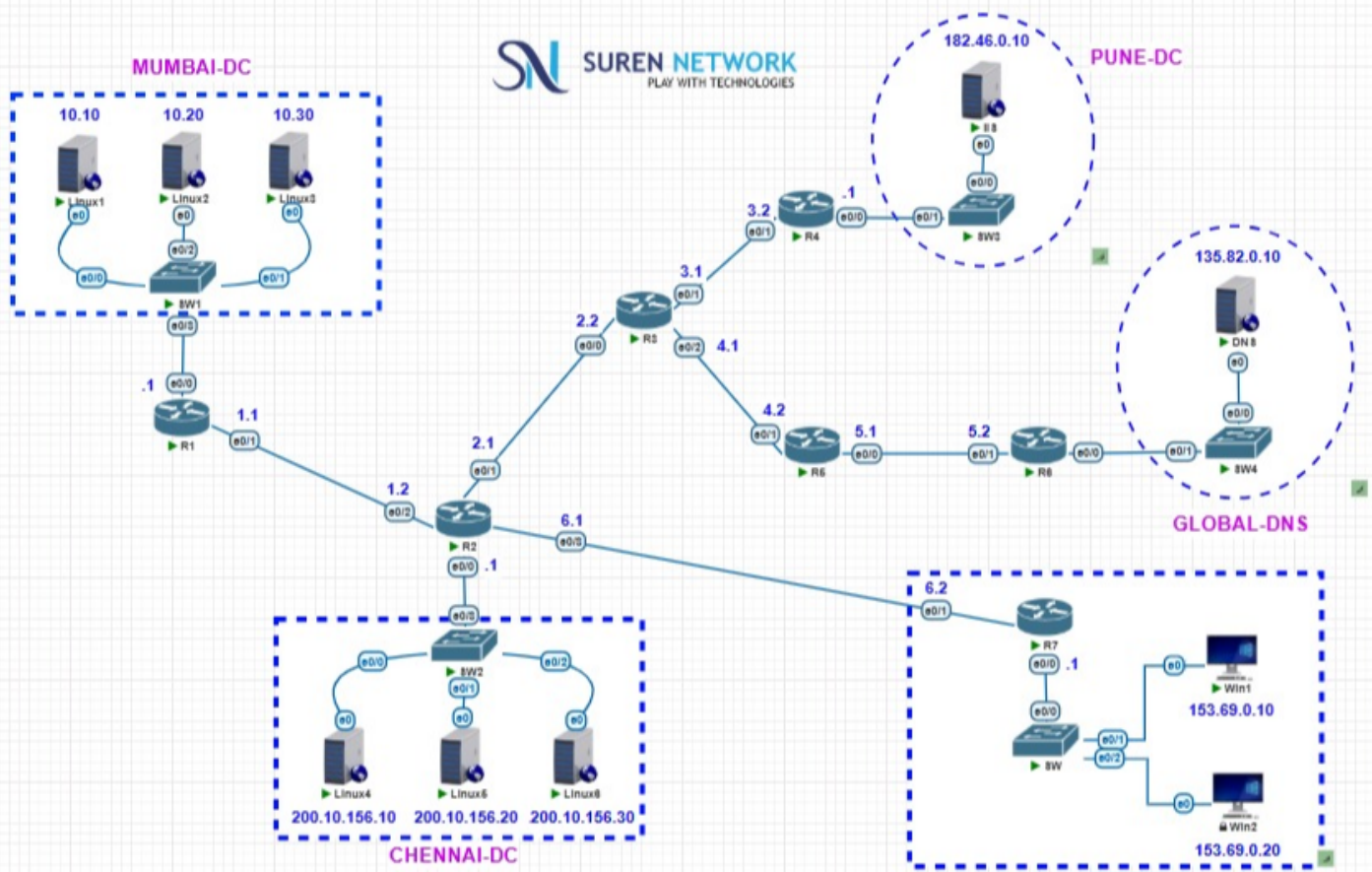




```
QEMU (Linux1)
route: ioctl 0x890c failed: No such process
Removing default gateway for Ethernet interface ..... [FAILED]
shutting down Ethernet interface eth0 ..... [ OK ]
shutting down loopback interface lo ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]
ifconfig
eth0    Link encap:Ethernet  HWaddr 00:50:00:00:01:00
        inet addr:10.0.0.10  Bcast:10.255.255.255  Mask:255.0.0.0
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:63 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B)  TX bytes:20286 (19.8 KiB)

lo      Link encap:Local Loopback
        inet addr:127.0.0.1  Mask:255.0.0.0
        UP LOOPBACK RUNNING  MTU:16436  Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

QEMU (Linux2)

```
route: ioctl 0x890c failed: No such process
Removing default gateway for Ethernet interface ..... [FAILED]
shutting down Ethernet interface eth0 ..... [ OK ]
shutting down loopback interface lo ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]
ifconfig
eth0  Link encap:Ethernet  HWaddr 00:50:00:00:02:00
      inet addr:10.0.0.20  Bcast:10.255.255.255  Mask:255.0.0.0
      UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
      RX packets:0 errors:0 dropped:0 overruns:0 frame:0
      TX packets:30 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:1000
      RX bytes:0 (0.0 B)  TX bytes:9660 (9.4 KiB)

lo  Link encap:Local Loopback
     inet addr:127.0.0.1  Mask:255.0.0.0
     UP LOOPBACK RUNNING  MTU:16436  Metric:1
     RX packets:0 errors:0 dropped:0 overruns:0 frame:0
     TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
     collisions:0 txqueuelen:0
     RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

```
QEMU (Linux3)
route: ioctl 0x890c failed: No such process
Removing default gateway for Ethernet interface ..... [FAILED]
shutting down Ethernet interface eth0 ..... [ OK ]
shutting down loopback interface lo ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]
ifconfig
eth0  Link encap:Ethernet  HWaddr 00:50:00:00:06:00
      inet addr:10.0.0.30  Bcast:10.255.255.255  Mask:255.0.0.0
      UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
      RX packets:0 errors:0 dropped:0 overruns:0 frame:0
      TX packets:23 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:1000
      RX bytes:0 (0.0 B)  TX bytes:7406 (7.2 KiB)

lo    Link encap:Local Loopback
      inet addr:127.0.0.1  Mask:255.0.0.0
      UP LOOPBACK RUNNING  MTU:16436  Metric:1
      RX packets:0 errors:0 dropped:0 overruns:0 frame:0
      TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:0
      RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

QEMU (Linux4)

```
route: ioctl 0x890c failed: No such process
Removing default gateway for Ethernet interface ..... [FAILED]
shutting down Ethernet interface eth0 ..... [ OK ]
shutting down loopback interface lo ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:03:00
          inet addr:200.10.156.10  Bcast:200.10.156.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:24 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:7728 (7.5 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

```
QEMU (Linux5)
route: ioctl 0x890c failed: No such process
Removing default gateway for Ethernet interface ..... [FAILED]
shutting down Ethernet interface eth0 ..... [ OK ]
shutting down loopback interface lo ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:04:00
          inet addr:200.10.156.20  Bcast:200.10.156.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:24 errors:0 dropped:0 overruns:0 frame:0
          TX packets:27 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:7728 (7.5 KiB)  TX bytes:8694 (8.4 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

```
QEMU (Linux6)
route: ioctl 0x890c failed: No such process
Removing default gateway for Ethernet interface ..... [FAILED]
shutting down Ethernet interface eth0 ..... [ OK ]
shutting down loopback interface lo ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:05:00
          inet addr:200.10.156.30  Bcast:200.10.156.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:37 errors:0 dropped:0 overruns:0 frame:0
          TX packets:20 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:11914 (11.6 KiB)  TX bytes:6440 (6.2 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

R1

```
R1#show
*Aug  6 08:19:52.588: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0              10.0.0.1        YES manual  up          up
Ethernet0/1              1.0.0.1         YES manual  up          up
Ethernet0/2              unassigned      YES TFTP    up          up
Ethernet0/3              unassigned      YES TFTP    up          up

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
*Aug  6 08:20:51.897: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#ip route 0.0.0.0 0.0.0.0 1.0.0.2
R1(config)#exit
R1#
*Aug  6 08:22:27.639: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 1.0.0.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 1.0.0.2
      1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      1.0.0.0/8 is directly connected, Ethernet0/1
L      1.0.0.1/32 is directly connected, Ethernet0/1
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.0.0.0/8 is directly connected, Ethernet0/0
L      10.0.0.1/32 is directly connected, Ethernet0/0
R1#
```

R2

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	200.10.156.1	YES	manual	up	up
Ethernet0/1	2.0.0.1	YES	manual	up	up
Ethernet0/2	1.0.0.2	YES	manual	up	up
Ethernet0/3	6.0.0.1	YES	manual	up	up

R2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 1.0.0.0/8 is directly connected, Ethernet0/2
L 1.0.0.2/32 is directly connected, Ethernet0/2
2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 2.0.0.0/8 is directly connected, Ethernet0/1
L 2.0.0.1/32 is directly connected, Ethernet0/1
S 3.0.0.0/8 [1/0] via 2.0.0.2
S 4.0.0.0/8 [1/0] via 2.0.0.2
S 5.0.0.0/8 [1/0] via 2.0.0.2
6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 6.0.0.0/8 is directly connected, Ethernet0/3
L 6.0.0.1/32 is directly connected, Ethernet0/3
S 10.0.0.0/8 [1/0] via 1.0.0.1
S 135.82.0.0/16 [1/0] via 2.0.0.2
S 153.69.0.0/16 [1/0] via 2.0.0.2
S 182.46.0.0/16 [1/0] via 2.0.0.2
200.10.156.0/24 is variably subnetted, 2 subnets, 2 masks
C 200.10.156.0/24 is directly connected, Ethernet0/0
L 200.10.156.1/32 is directly connected, Ethernet0/0

R3

```
R3(config)#ip route 182.46.0.0 255.255.0.0 3.0.0.2
R3(config)#ip route 135.82.0.0 255.255.0.0 4.0.0.2
R3(config)#ip route 153.69.0.0 255.255.0.0 2.0.0.1
R3(config)#exit
R3#s
*Aug  6 08:54:30.400: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              2.0.0.2         YES manual up          up
Ethernet0/1              3.0.0.1         YES manual up          up
Ethernet0/2              4.0.0.1         YES manual up          up
Ethernet0/3              unassigned      YES TFTP  up          up
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 2.0.0.1
    2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    2.0.0.0/8 is directly connected, Ethernet0/0
L    2.0.0.2/32 is directly connected, Ethernet0/0
    3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    3.0.0.0/8 is directly connected, Ethernet0/1
L    3.0.0.1/32 is directly connected, Ethernet0/1
    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    4.0.0.0/8 is directly connected, Ethernet0/2
L    4.0.0.1/32 is directly connected, Ethernet0/2
S    5.0.0.0/8 [1/0] via 4.0.0.2
S    6.0.0.0/8 [1/0] via 2.0.0.1
S    10.0.0.0/8 [1/0] via 2.0.0.1
S    135.82.0.0/16 [1/0] via 4.0.0.2
S    153.69.0.0/16 [1/0] via 2.0.0.1
S    182.46.0.0/16 [1/0] via 3.0.0.2
S    200.10.156.0/24 [1/0] via 2.0.0.1
R3#
```

 R4

```
R4(config-if)#exit
R4(config)#interface ethernet0/1
R4(config-if)#ip address 3.0.0.2 255.0.0.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#ip route 0.0.0.0 0.0.0.0 3.0.0.1
R4(config)#exit
R4#aho
*Aug  6 08:58:42.826: %SYS-5-CONFIG_I: Configured from console by console
R4#show ip interface brief
Interface                               IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0                             182.46.0.1      YES manual  up          up
Ethernet0/1                             3.0.0.2         YES manual  up          up
Ethernet0/2                             unassigned      YES TFTP   up          up
Ethernet0/3                             unassigned      YES TFTP   up          up

R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 3.0.0.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 3.0.0.1
      3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      3.0.0.0/8 is directly connected, Ethernet0/1
L      3.0.0.2/32 is directly connected, Ethernet0/1
      182.46.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      182.46.0.0/16 is directly connected, Ethernet0/0
L      182.46.0.1/32 is directly connected, Ethernet0/0
R4#
```

 R5

```
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#ip route 1.0.0.0 255.0.0.0 4.0.0.1
R5(config)#ip route 2.0.0.0 255.0.0.0 4.0.0.1
R5(config)#ip route 3.0.0.0 255.0.0.0 4.0.0.1
R5(config)#ip route 6.0.0.0 255.0.0.0 4.0.0.1
R5(config)#ip route 10.0.0.0 255.0.0.0 4.0.0.1
R5(config)#ip route 200.10.156.0 255.255.255.0 4.0.0.1
R5(config)#ip route 182.46.0.0 255.255.0.0 4.0.0.1
R5(config)#ip route 135.82.0.0 255.255.0.0 5.0.0.2
R5(config)#ip route 153.69.0.0 255.255.0.0 4.0.0.1
R5(config)#exit
R5#
*Aug  6 09:10:37.162: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 4.0.0.1
S    2.0.0.0/8 [1/0] via 4.0.0.1
S    3.0.0.0/8 [1/0] via 4.0.0.1
S    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    4.0.0.0/8 is directly connected, Ethernet0/1
L    4.0.0.2/32 is directly connected, Ethernet0/1
S    5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    5.0.0.0/8 is directly connected, Ethernet0/0
L    5.0.0.1/32 is directly connected, Ethernet0/0
S    6.0.0.0/8 [1/0] via 4.0.0.1
S    10.0.0.0/8 [1/0] via 4.0.0.1
S    135.82.0.0/16 [1/0] via 5.0.0.2
S    153.69.0.0/16 [1/0] via 4.0.0.1
S    182.46.0.0/16 [1/0] via 4.0.0.1
S    200.10.156.0/24 [1/0] via 4.0.0.1
R5#
```

 R6

```
R6(config-if)#ip address 5.0.0.2 255.0.0.0
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#ip route 0.0.0.0 0.0.0.0 5.0.0.0.
*Aug  6 09:14:04.032: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R6(config)#ip route 0.0.0.0 0.0.0.0 5.0.0.1
R6(config)#exit
R6#s
*Aug  6 09:14:12.194: %SYS-5-CONFIG_I: Configured from console by console
R6#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0              135.82.0.1      YES manual  up          up
Ethernet0/1              5.0.0.2         YES manual  up          up
Ethernet0/2              unassigned      YES TFTP   up          up
Ethernet0/3              unassigned      YES TFTP   up          up

R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 5.0.0.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 5.0.0.1
      5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      5.0.0.0/8 is directly connected, Ethernet0/1
L      5.0.0.2/32 is directly connected, Ethernet0/1
      135.82.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      135.82.0.0/16 is directly connected, Ethernet0/0
L      135.82.0.1/32 is directly connected, Ethernet0/0
R6#
```

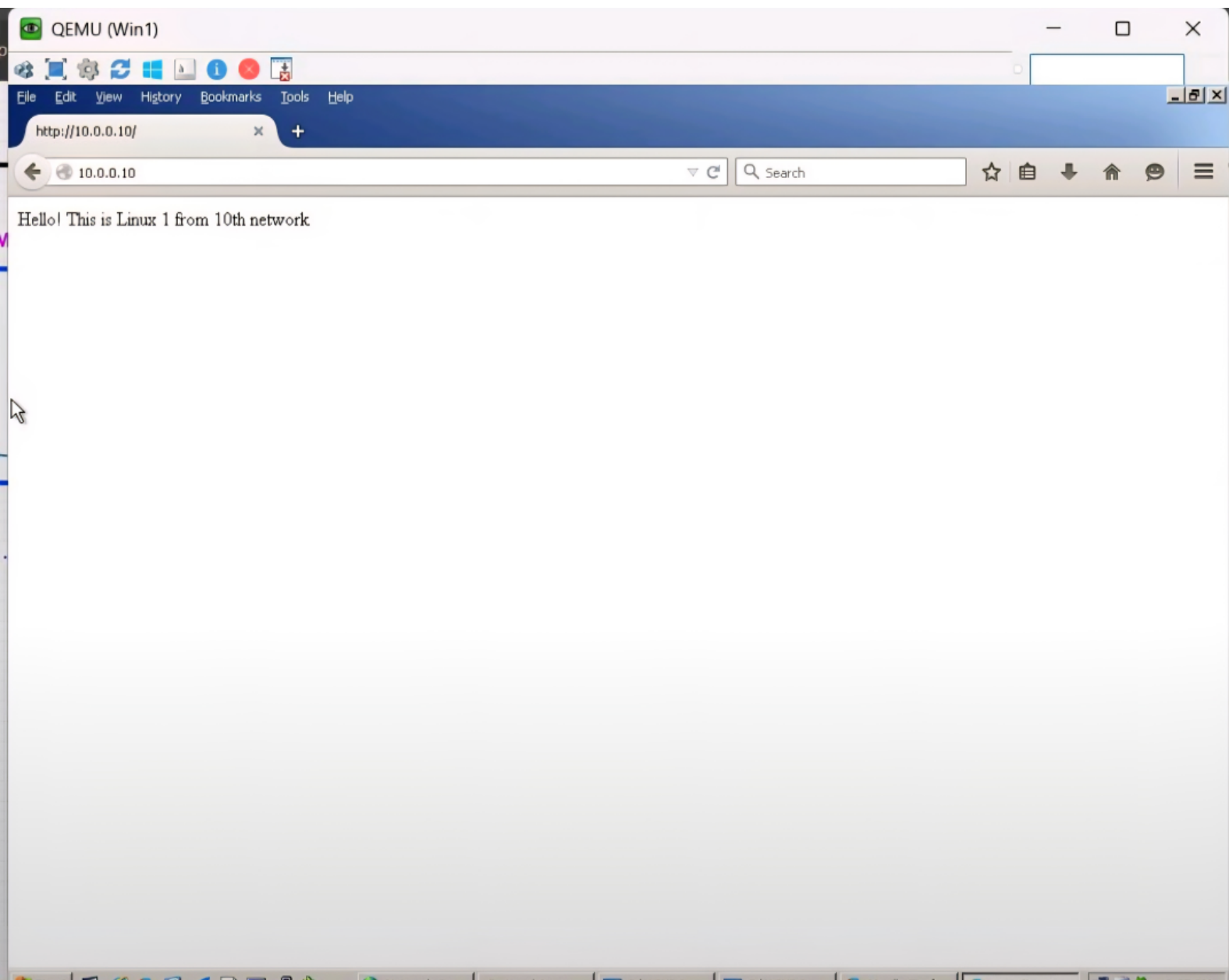
R7

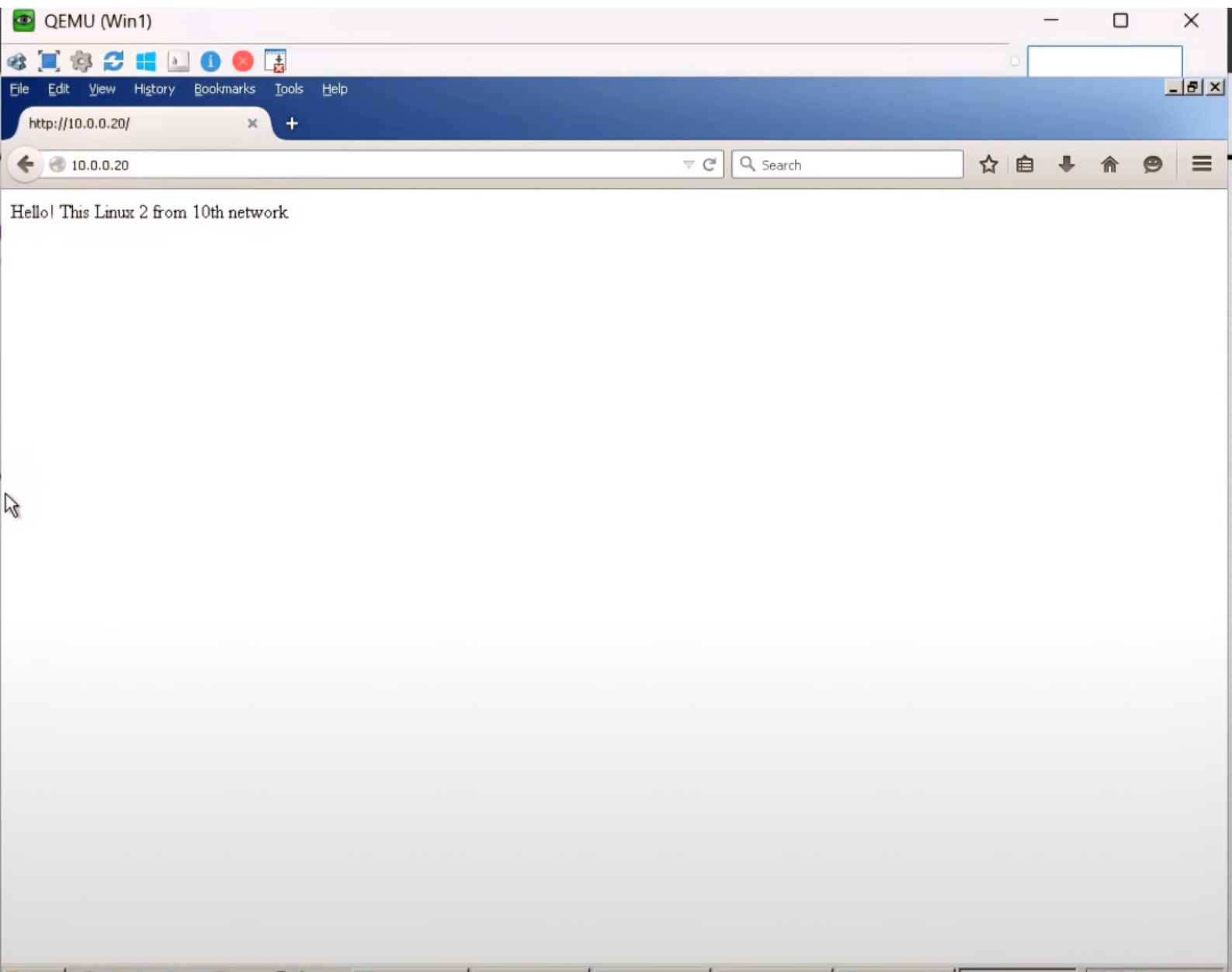
```
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#interface ethernet0/1
R7(config-if)#ip address 6.0.0.2 255.0.0.0
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#ip route 0.0.0.0 0.0.0.0 6.0.0.1
R7(config)#exit
R7#sh
*Aug 6 10:52:13.576: %SYS-5-CONFIG_I: Configured from console by console
R7#show ip interface brief
Interface                IP-Address      OK? Method Status    Prot
oocol
Ethernet0/0              153.69.0.1     YES manual up        up
Ethernet0/1              6.0.0.2        YES manual up        up
Ethernet0/2              unassigned     YES TFTP  up        up
Ethernet0/3              unassigned     YES TFTP  up        up

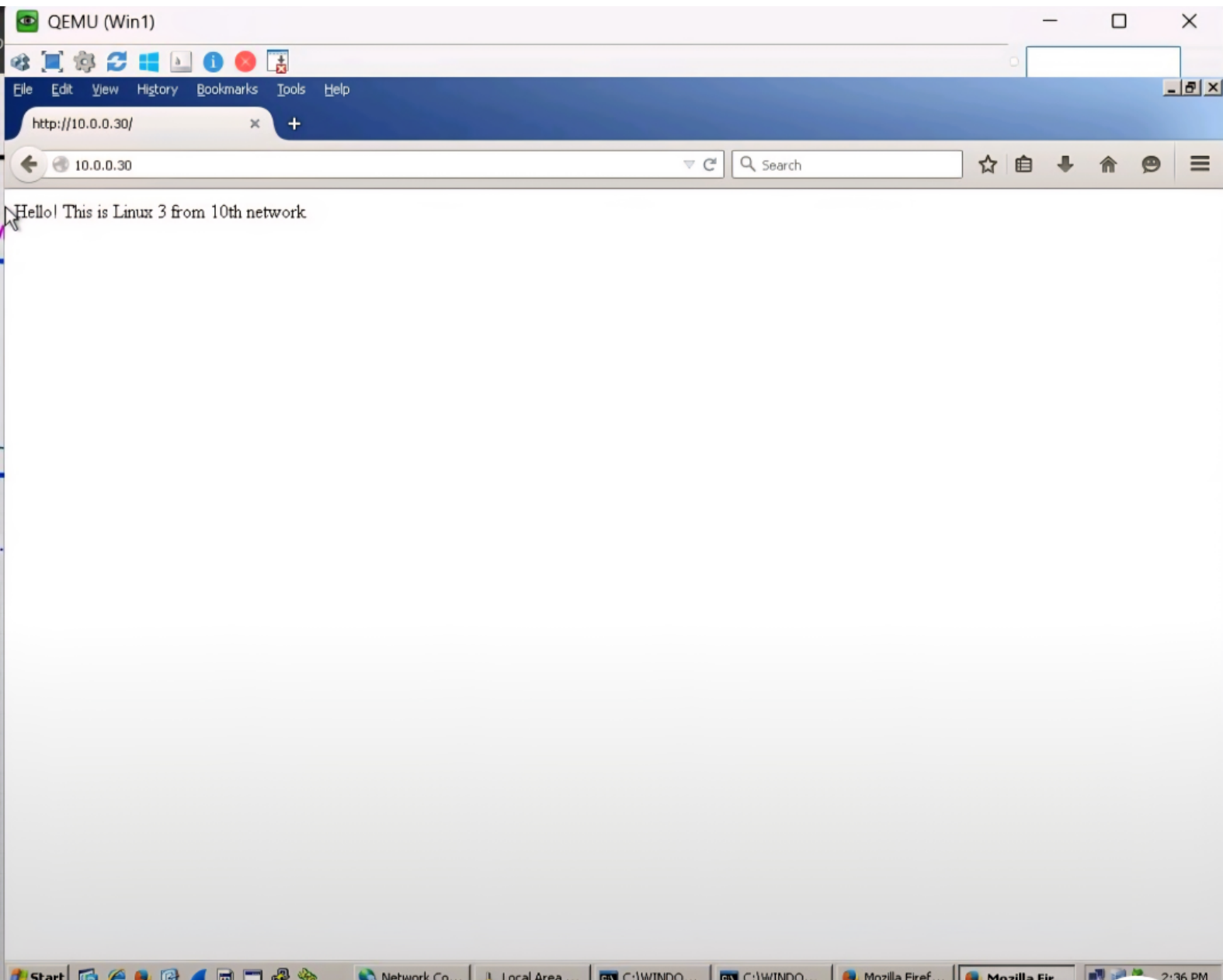
R7#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 6.0.0.1 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 6.0.0.1
     6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     6.0.0.0/8 is directly connected, Ethernet0/1
L     6.0.0.2/32 is directly connected, Ethernet0/1
     153.69.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     153.69.0.0/16 is directly connected, Ethernet0/0
L     153.69.0.1/32 is directly connected, Ethernet0/0
R7#
```







QEMU (DNS)

DNS Manager

File Action View Help

DNS

SERVER2016

Forward Lookup Zones

in

linux1

linux2

linux3

linux4

linux5

linux6

com

windows

Reverse Lookup Zones

Trust Points

Conditional Forwarders

Name	Type	Data	Timestamp
www	Host (A)	182.46.0.10	

1:29 PM

7/25/2026

